QUESTION BANK

FUNDAMENTALS OF QUANTUM COMPUTING

UNIT I

Short Question & Answers:

1.Define Rayleigh-jeans law?

Rayleigh-Jean's law:
According to Rayleigh-jean's the radiation emitted by the black body at a temperature is directly proportional to absolute temperature of black body and inversely proportional to fourth power of the wavelength.

$$E \propto T$$

$$E(u) \propto \frac{1}{\lambda^4}$$

$$E(u) = \frac{8\pi kT}{\lambda^4}$$

2.Define Stephen-Boltzmann law?

Stephen - Boltzmann law:
Stephen-Boltzmann law is the relation between the radiation emitted by the black body to the temperature.

- The total radiation emitted by the black body is directly proportional to the fourth power of absolute temperature of black body.

i.e., E α T4
E = σT4
σ = 5.67 x 10 -8 w/m 2 – K4 .

3.Write the expression for the Weins law?

Weins law:
It explains a shifting of peak wavelength with the temperature.
The peak wavelength emitted by the black body a temperature is inversely proportional to the absolute temperature of black body.

$$i.e., \lambda_m \, \alpha \frac{1}{T}.$$

$$\lambda_m = 2.87 \times 10^{-3} \frac{mK}{T}.$$

4.Define black body and black body radiation?

Black body:

- A perfect black body one which observed and emit all the wavelength of incident radiation on it

Black body radiation:

- The radiation emitted by the black body is known as Black body radiation.

5.Define Matter waves?

Matter wave:

Any moving particle is associated with wave nature those waves are called matter waves.

6.Define photoelectric effect

Photo electric effect:

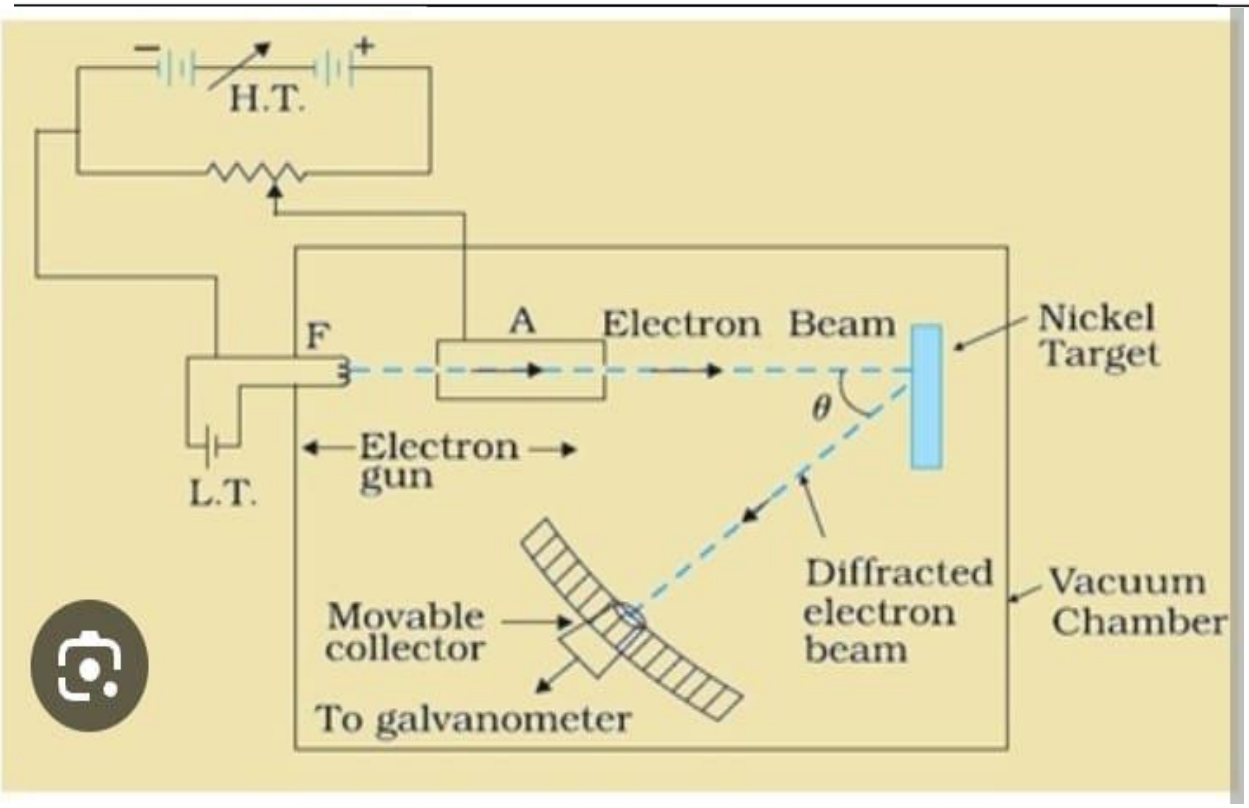
When the electromagnetic radiation (light) is incident on a photo sensitive metal if the electrons are emitted/ejected from the surface of metal then that phenomenon is called photo electric effect.

LONG QUESTIONS & ANSWERS:

1.Describe the experiment to verify the existence of matter waves?
(OR)
Explain Davisson-Germer Experiment ?

Davisson-Germer Experiment:

- The experimental set up of Davisson and Germer experiment is.



Davisson-Germer experiment consist of three parts

ElectronGun:

It consists of a filament, low-tension battery, high-tension battery and pin-hole arrangement. When the filament is heated, electrons are emitted and accelerated by high voltage to form a fine electron beam.

NickelCrystal(Target):

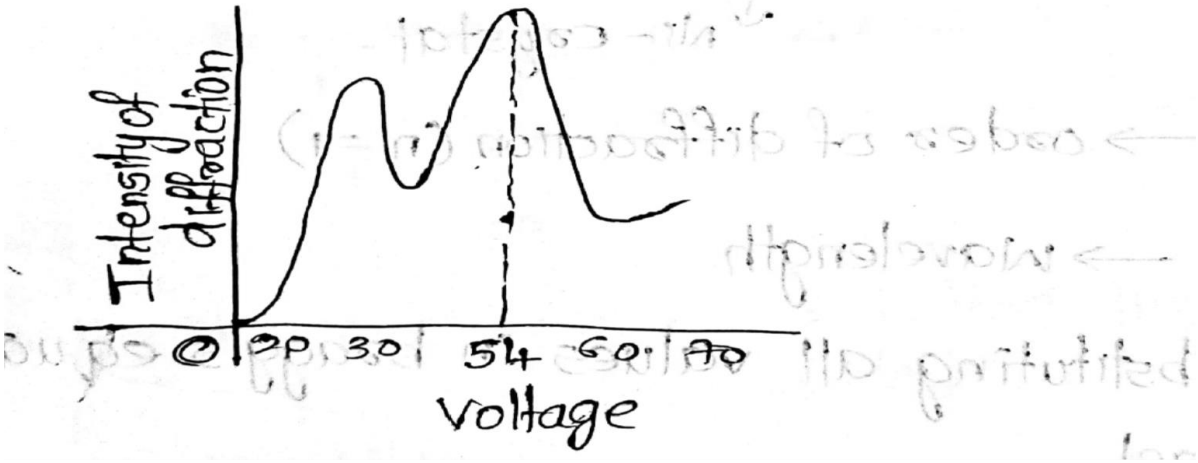
The high-speed electron beam is made incident on the Ni-crystal. Due to the regularly arranged atomic planes, electrons are diffracted in all directions.

CircularScaleArrangement:

It has a movable electron collector mounted on a circular scale and connected to a sensitive galvanometer. It collects the diffracted electrons and measures their intensity at different angles.

Calculation of wavelength associate from the electron:

When the voltage from High tension battery V = 54, The angle between incident beam to diffracted beam is 50° then we will get maximum diffraction pattern (the order of diffraction n=1).



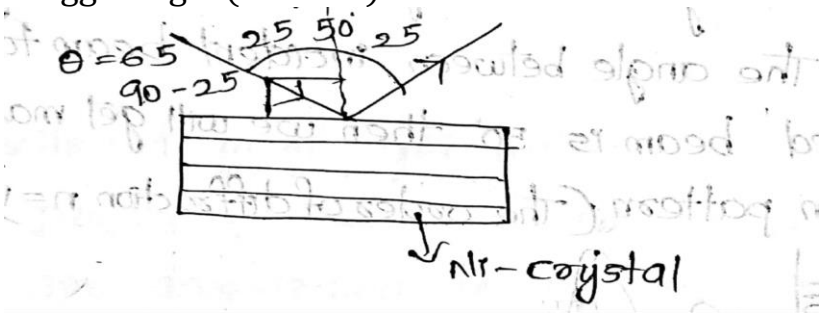
The wavelength associate from the electron according to de-broglie

$$\lambda = \frac{12.26}{\sqrt{V}} A°$$

$$\lambda = \frac{12.26}{\sqrt{54}}$$

$$\lambda = 1.65A°$$

The wavelength associate from the Bragg's law for the electron.
Bragg's law
$2dsin\theta = n\lambda$
d ➜ Inter planar distance for the Ni-crystal.
$d = 0.91°$
That can calculate by the (X-ray diffraction method)
Bragg's angle ($\theta = 65°$)



n➜Order of diffraction.
λ➜Wavelength.

By substituting all values in Bragg's equation we get
$= 2 \times 0.91 \times sin65 = 1 \times \lambda$
$\lambda = 1.66A°$
A good agreement between De-Broglie and Bragg's law for the electron
Hence, the wave nature of electron was proved.

2.Explain the Heisenberg uncertainty principle?

Heisenberg uncertainty principle:

- According to Heisenberg it is difficult to find the position and momentum simultaneously.

- x is the exact position of a particle 'p' is the exact momentum of a particle.
- Δx is the error to find the position of a particle, ΔP is the error to find the momentum of a particle. Then the relation between Δx *and* ΔP from the Heisenberg can be written as

$$\Delta x \cdot \Delta P \geq \frac{h}{4\pi} \; or \; \Delta x \cdot \Delta P \geq \frac{h}{2}$$

$$\therefore h = \frac{h}{2\pi}$$

Case(i) if $\Delta x = 0$ (no error to find the position of particle)

$$\Delta x \cdot \Delta P \geq \frac{h}{4\pi}$$

$$\Delta P \geq \frac{h}{4\pi \cdot \Delta x}$$

$$\Delta P \geq \frac{K}{4\pi(0)}$$

$$\Delta P = \infty$$

i.e., cannot identify the momentum of the particle.

case(ii) $\Delta P = 0$ (no error to find momentum of the particle)

$$= \Delta x \cdot \Delta P \geq \frac{h}{4\pi}$$

$$\Delta x \geq \frac{h}{4\pi(\Delta P)}$$

$$\Delta x = \infty$$

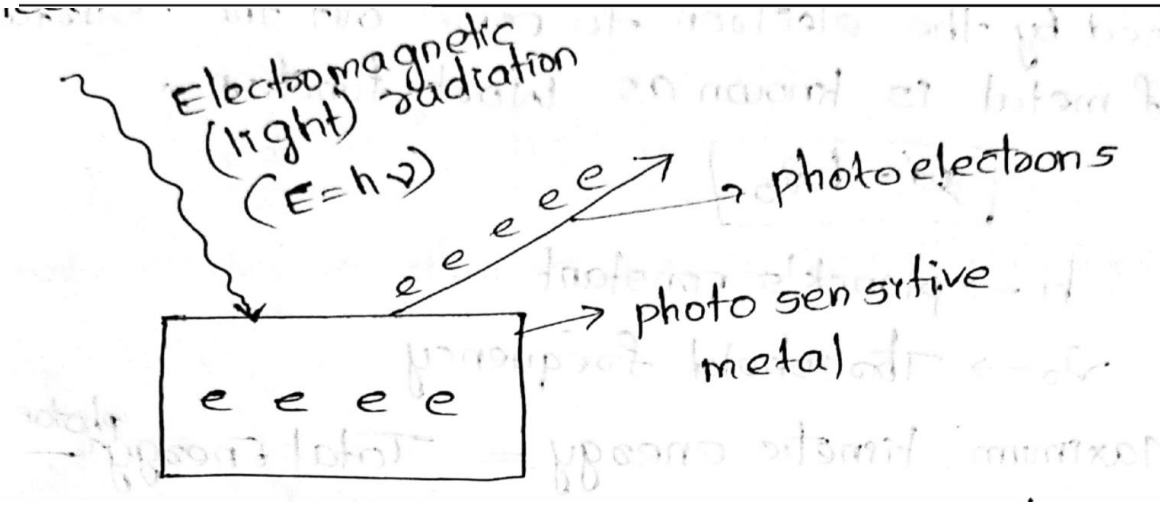
i.e., cannot identify the position of the particle similar relation exist between energy and time.

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \; or \; \Delta E \cdot \Delta t \geq \frac{h}{2}$$

3.Define photoelectric effect? With mathematical expression?

Photo electric effect:

When the electromagnetic radiation (light) is incident on a photo sensitive metal if the electrons are emitted/ejected from the surface of metal then that phenomenon is called photo electric effect.



Einstein explained photo electric effect phenomena by using Max-Planck quantum theory.
According to Einstein energy radiated in the form of photon.
Energy of photon E = hυ
Explanation:
When the electromagnetic radiation is incident on a photosensitive metal photon (E = hυ) collides with the electron, during this collision photon transfer all the amount of energy to electron.
In these total energy part of energy used by the electron to come out the surface of the metal remaining part of energy used by the electron as maximum Kinetic energy with this electron leaves from the metal surface.
Work function:
The minimum amount of energy used by the electron to come out the surface of metal is known as work function.

$\phi = h\upsilon_0$

h →Plank's constant.
υ0 →Threshold frequency.

Maximum Kinetic energy = Total energy photon – Work function.

$KE = h\upsilon - \phi$

$\frac{1}{2}mv^2 = h\upsilon - h\upsilon_0$

$\frac{1}{2}mv^2 = h(\upsilon - \upsilon_0)$

m →Mass of the electron.
v →Velocity of electron.
υ→Incident radiation frequency.
υ_0→Threshold frequency

Condition for photo electric effect:
We know mathematical expression for photo electric effect.

$$\frac{1}{2}mv^2 = h\upsilon - h\upsilon_0$$

Kinetic energy is either positive (or) zero.

i.e., K.E ≥ 0
$\rightarrow hv - hv_0 \geq 0$

$\hbar\ v \geq \hbar\ \ v_0$
$v \geq v_0$
Conclusion:
To observe the photo electric effect phenomena incident radiation frequency must be greater than (or) equal to threshold frequency.

4.Derive the De- Broglie wave length for moving electron ?

De-Broglie wavelength for the electron:
Let us consider an electron with charge `e' and mass 'm' moving with velocity 'υ' applied potential `V'
The energy gained by the electron due to potential 'V'
E = ev ---------- ❶
The energy gained by the electron due to motion
$E = \frac{1}{2}\ mv^2$ ⟶ ❷
From 1 and 2
$$ev = \frac{1}{2}mv^2$$
$$mv^2 = 2ev$$
Multiplying with 'm' on both sides
$$m \times mv^2 = m2ev$$
$$m^2 \times v^2 = 2mev \qquad (m \times v = p)$$
$$p^2 = 2mev$$
$$p = \sqrt{2mev}$$
$$we\ know\ that\ \lambda = \frac{h}{p}$$
$$\lambda = \frac{h}{\sqrt{2mev}}$$
h - Plank's constant 6.625 × 10 -34 J. sec.
m - Mass of e— 9.1 × 10 -31 Kg.
e – Charge of an e— 1.602 × 10 -19 columb.
V – Potential difference.
$$\lambda = \frac{6.625\ \times\ 10^{-34}\ J.sec}{\sqrt{2 \times 9.1 \times 10^{-31}\ Kg \times 1.602 \times\ 10^{-19} V}}$$

$$\lambda = \frac{12.26}{\sqrt{V}}\ A°$$

5.Derive the time- independent Schrodinger's wave equation of electron ?

Let us consider an electron with mass 'm' moving with velocity 'v' in positive 'x' direction under the potential 'V'.

The general wave equation for the electron can be represented as

$$\psi(x) = A\,sin\left(\frac{2\pi}{\lambda}\right)x \quad ---------- \quad (1)$$

Differentiate above with r to "x".

$$\frac{d\psi(x)}{dx} = A\,cos\left(\frac{2\pi}{\lambda}\right)x \times \frac{2\pi}{\lambda}$$

gain differentiate above with respect to "x"

$$\frac{d^2\psi(x)}{dx^2}$$

$$= -A\,sin\left(\frac{2\pi}{\lambda}\right)x \cdot \left(\frac{2\pi}{\lambda}\right)\left(\frac{2\pi}{\lambda}\right)$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{4\pi^2}{\lambda^2}A\,sin\left(\frac{2\pi}{\lambda}\right)x$$

From equation 1

$$\frac{d^2\psi(x)}{dx^2} = -\frac{4\pi^2}{\lambda^2}\psi(x)$$

$$= \frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\lambda^2}\psi(x) = 0 \quad ---------- \quad (2)$$

From the De-Broglie

$$\therefore \lambda = \frac{h}{mv}$$

λ substitute in equation 2

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\left(\frac{h}{mv}\right)^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2}\psi(x) = 0 \quad ---------- \quad (3)$$

Total energy of particle = K.E + P.E.

$$E = \frac{1}{2}mv^2 + V$$

$$E - V = \frac{1}{2}mv^2$$

$$2\,(E - V) = mv^2.$$

Multiply with "m" on both sidesħ

$$m \times 2(E - V) = m \times mV^2$$

$$2m(E - V) = m^2v^2 \;\; substitute\ in\ \ eqn\ 3$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2\big(2m(E - V)\big)}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m(E-V)}{h^2}\psi(x) = 0 \qquad \text{----------} \quad (4)$$

$\psi(x)$ ➔Wave function.

x ➔ Propagation of wave.

m ➔Mass of the particle.

E ➔Total Energy of the particle.

V ➔Applied potential.

h ➔Planck's constant.

Equation 4 is known as Schrodinger wave equation (or) Schrodinger 1-D wave equation (or) Time independent wave equation.

6. Derive the time- dependent Schrodinger's wave equation of electron?

Schrodinger's time dependent wave equation.

W.K.T $\nabla^2\psi + 2m/\hbar^2 [E-V]\psi = 0$

Multiply on both sides $\hbar^2/2m$

$\hbar^2/2m\,[\nabla^2\psi + 2m/\hbar^2 [E-V]\psi] = 0 \times \hbar^2/2m$

$\frac{\hbar^2}{2m} \times \nabla^2\psi + \frac{\hbar^2}{2m} \times \frac{2m}{\hbar^2}[E-V]\psi = 0 \times \frac{\hbar^2}{2m}$

$\frac{\hbar^2}{2m} \times \nabla^2\psi + [E-V]\psi = 0$

$E\psi = \frac{-\hbar^2}{2m} \times \nabla^2\psi + V\psi$

$E\psi = [-\frac{\hbar^2}{2m} \times \nabla^2 + V]\psi$

$E\psi = H\psi \qquad [H = \frac{-\hbar^2}{2m} \times \nabla^2 + V]$

H is called Hamiltonian operator. E is called energy operator. This is known as Schroedinger's time dependent wave equation.

Note:- The Schrodinger wave equation describes the behaviour of matter waves.

7.Write a note on Born's interpretation of wave function?

- Wave function is a mathematical expression represented in complex

number. It has no direct physical meaning.

- square of the wave function $[\psi^2(x) = \psi(x)\psi(x)]$

  is called probability density
- It gives finding the position of a particle in a given region.

If the particle is presented in a given region then probability density maximum this condition is known as normalization Condition.

Mathematically we can represent as.

$$\int \psi^2(x)d(x) = 1 \qquad [in\ 1-D]$$
$$\int \psi^2(x,y,z)dv = 1 \qquad [in\ 3-D]$$

If the particle is not existed in a given region, then probability density minimum this condition is known as orthogonal condition.

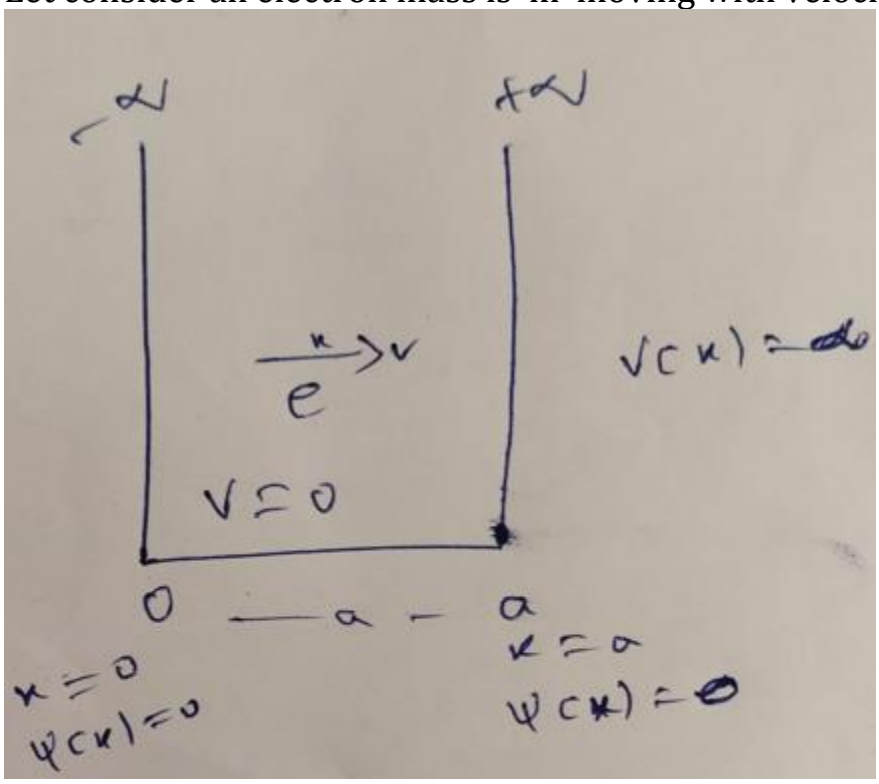
Mathematically represented as.

$$\int \psi^2(x)dx = 0 \qquad [in\ 1-D]$$
$$\int \psi^2(x,y,z)dv = 0 \qquad [in\ 3-D]$$

8.Derive a energy equation for electron by Trapping in one dimensional potential box?

Let us consider a one dimensional potential box of width 'a'

Let consider an electron mass is 'm' moving with velocity v in positive x direction.



The motion of microscopic particle is explained by Schrodinger wave equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m\big(E - V(x)\big)}{h^2}\psi(x) = 0 \qquad ---------- \quad (1)$$

Potential :
Inside of the box
$V(x) = 0\,; 0 < x < a$
Outside of box
$V(x) = \infty\,; x < 0\ (or) x\ > a$
Apply Schrodinger wave equation inside box of

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m\big(E - V(x)\big)}{h^2}\psi(x) = 0 \qquad [\therefore\ V(x) = 0]$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m(E - 0)}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{8\pi^2 mE}{h^2}\right)\psi(x) = 0 \qquad [\therefore K^2 = \frac{8\pi^2 mE}{h^2}]$$

$$\frac{d^2\psi(x)}{dx^2} + K^2\psi(x) = 0 \qquad ---------- \ (2)$$

The solution of equation 2 from the second order differentiation can be written as
$\psi(x) = A\ sinkx + B\ coskx$ ---------- (3)

Boundary conditions for 1-D potential box.
At I-Boundary the wave function is 0

$$At\ x = 0\ \Longrightarrow \psi(x) = 0$$

At II-Boundary the wave function 0

$$At\ x = a\ \Rightarrow \psi(x) = 0$$

Apply Boundary condition I & II for equation 3
$At\ x = 0\ \Rightarrow \psi(x) = 0$
$\psi(\alpha) = A\ sinkx + B\ Coskx$
$0 = A\ sink(0) + B\cos(0)$
$\Rightarrow 0 + B = 0$
$B = 0$
$At\ x = a \Rightarrow \psi(x) = 0$
$0 = A\ sink(a) + B\ cosk(a)$
$= A\ sinka + Bcoska \qquad (\therefore B = 0)$
$\Rightarrow A\ sinka = 0$
$Sinka = 0$
$ka = \sin^{-1}(0)$
$= Sin^{-1}\ (Sin\pi)$
$ka = n\pi.$
$k = \frac{n\pi}{a} \qquad [\ k \rightarrow wave\ vector]$
$\therefore k^2 = \frac{8\pi^2 mE}{h^2}$
Substitute “k” value.

$$\left(\frac{n\pi}{a}\right)^2 = \frac{8\pi^2 mE}{h^2}$$

$$\frac{n^2\pi}{a^2} = \frac{8\pi^2 mE}{h^2}$$

$$E_n = \frac{n^2h^2}{8ma^2}$$

n ➜No. of Quantum energy.
E ➜Energy of electron.
h➜Planck's constant.
m➜Mass of electron.
a➜Width of box.

If n=1

$$E_1 = \frac{(1)^2h^2}{8ma^2} = \frac{h^2}{8ma^2} \qquad (Ground\ state\ of\ electron)$$

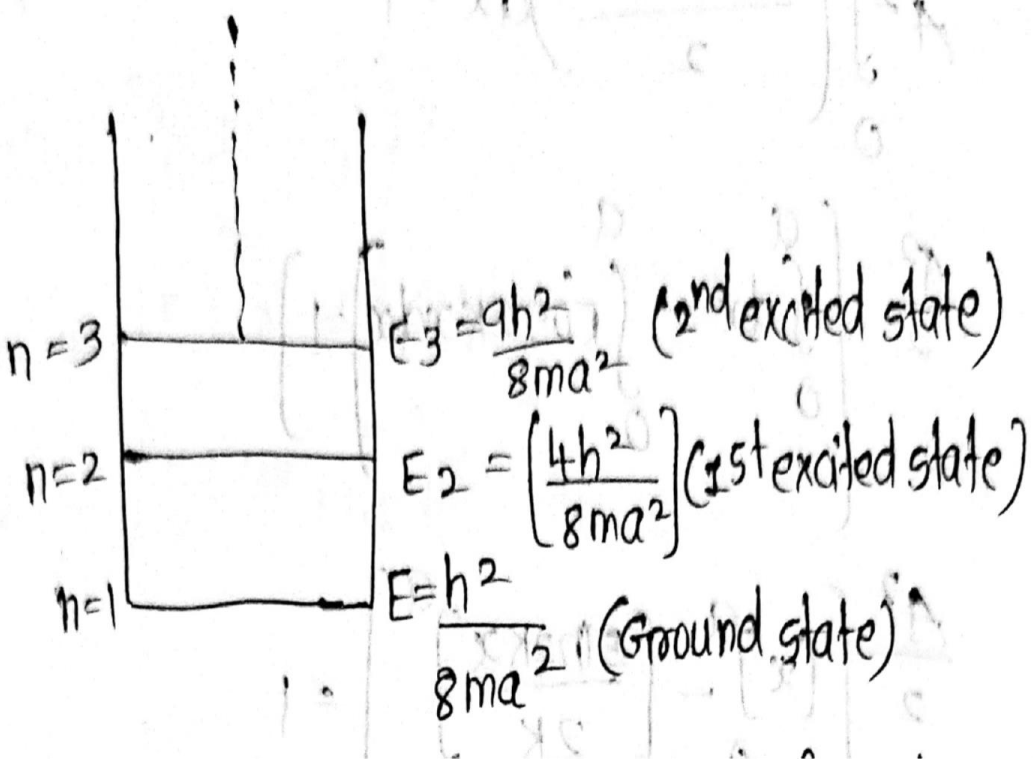
If n=2

$$E_2 = \frac{4h^2}{8ma^2} \qquad (1st\ excited\ state)$$

If n=3

$$E_3 = \frac{9h^2}{8ma^2} \qquad (2nd\ excited\ state)$$

Energy of electron is Quantized and emit in discrete manner.



UNIT II

Short Question & Answers:

1.Write any two assumptions given by Sommerfield?

1) Valence electrons move free freely in constant potential with in the boundary of metal. Hence, electron trapped in potential box.
2) The distribution of electron in various energy level occurs as per Pauli's exclusion principle.

2.Write any two Merits of Sommerfield theory?

1.It explains electrical conductivity, thermal conductivity of metal.
2.It explains the variation of electrical conductivity & thermal with temperature.

3.Write any two Merits of classical free electron theory?

1.It explained Ohm's law.
2.It explained electrical conductivity, thermal conductivity off metal.
It explained Wiedemann - Franz law

4.Explain about the Insulators with examples?

An insulator is a material that does not allow electric current to flow easily because it has very few or no free electrons.
Example: Glass, Rubber

5.Define Semiconductors with two examples?

A semiconductor is a material whose electrical conductivity lies between conductors and insulators. Its conductivity can be controlled by temperature or doping.
Example: Silicon, Germanium

6.Define conductors with two examples?

A conductor is a material that allows electric current to flow easily because it has a large number of free electrons.
Example: Copper, Aluminium

LONG QUESTIONS & ANSWERS:

1.What are the assumptions, merits and Demerits of classical free electron theory?

Classical free electron theory:

Assumptions:

- In Metal large no. of free electrons moves freely in all possible directions.
- These electrons behave as gas molecules in container which obeys kinetic theory of gases.

- In the absence of field, the energy associate with electron $E = \frac{3}{2}KT$

  which is equal to kinetic energy

$$\frac{3}{2}KT = \frac{1}{2}mv_{th}^2$$

Vth = Thermal velocity.

- Metal consist of positive ion core in fixed position and electron move freely collides with positive ion cove (or) free electron (or) with the metal boundary.
- These collisions are perfectly elastic collision. Hence, conductivity is due to free electrons.
- The electron velocity obeys Maxwell-Distribution velocity.
- The potential energy of electron is always constant value.
- If electric field is applied the electron gets accelerated, the velocity of electron opposite direction of electric field this velocity is called drift velocity (Vd).

Merits:

- It explained Ohm's law.
- It explained electrical conductivity, thermal conductivity off metal.
- It explained Wiedemann - Franz law.
- It explained optical properties of metal.

De-Merits:

- It fails to explain electrical conductivity & thermal conductivity of semiconductors & Insulators.
- It fails to explain the variation of electrical conductivity, thermal conductivity at low temperature.
- It fails to explain photoelectric effect, black body radiation and specific heat of electron.
- It fails to explain the susceptibility of paramagnetic material and Ferromagnetic material based on temperature.

2.What are the assumptions, merits and Demerits of quantum free electron theory?

Quantum free electron theory:

Assumptions:

- It overcomes drawbacks of classical free electron theory.
- Sommerfeld explained them by using Fermi Dirac distribution instead of Maxwell-Boltzmann distribution.

1) Valence electrons move free freely in constant potential with in the boundary of metal. Hence, electron trapped in potential box.
2) The distribution of electron in various energy level occurs as per Pauli's exclusion principle.

3) Distribution of energy of electrons obeys fermi-dirac distribution.
4) The energy of electron is quantized & discrete.

$$i.e., E_n = \frac{n^2h^2}{8ma^2}$$

Merits:

- It explains electrical conductivity, thermal conductivity of metal.
- It explains the variation of electrical conductivity & thermal with temperature.
- It explains photoelectric effects, compton effect & black body radiation.
- It explains specific heat of electron & susceptibility of para magnetic & ferromagnetic material with the temperature.

De-merits:

- It fails to distinguish between conductors, insulators, semiconductor.
- It fails to explain Hall effect phenomenon.
- According to quantum free electron theory only two electrons exist in fermi energy level, which are responsible for conductivity but this is not true.

3. Derive the expression for the Bloch equation?

Bloch Theorem:

- let us consider a +ve ion core ('N' no of ions) with periodicity 'a' (distance between any two ions).
- Let consider an electron moving freely between the positive ion core, when the electron at the site of +ve ion the potential is minimum (V=0), when electrons between the +ve ions the potential is maximum (V0).

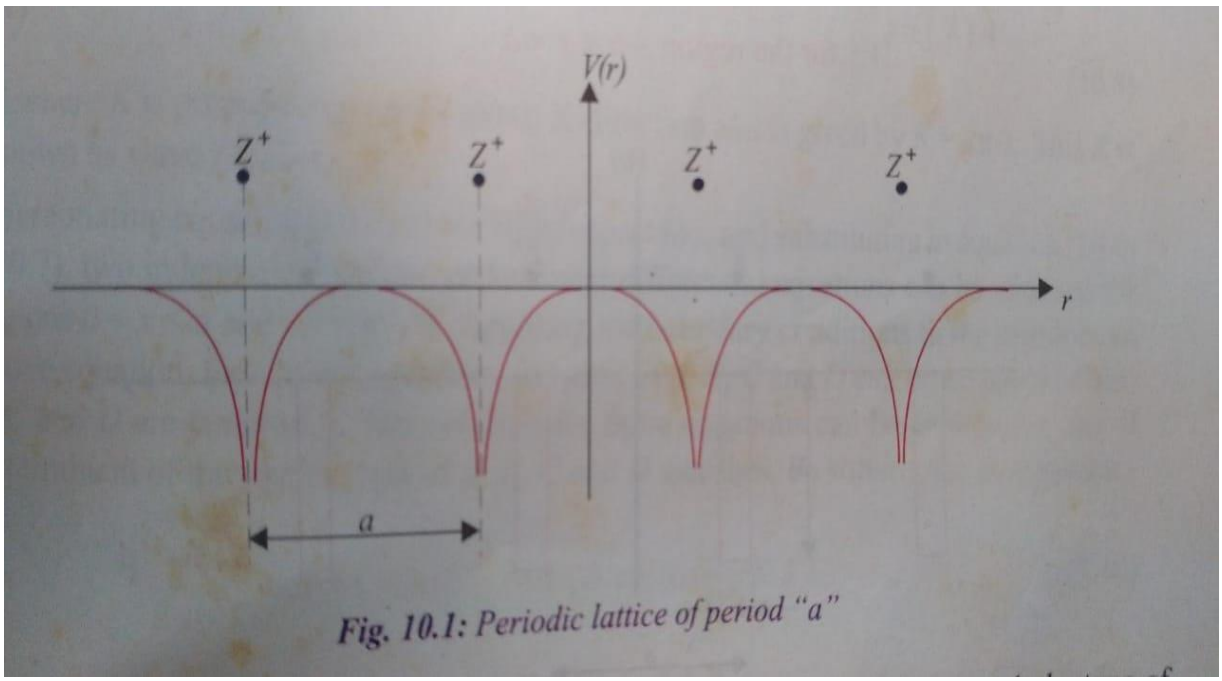


Fig. 10.1: Periodic lattice of period "a"

- The motion of electron is explained by schrodinger equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m[E - V(x)]}{h^2}\psi(x) = 0 \qquad ------ \quad (i)$$

The potential for single ion $V(x) = V(x+a)$ $\quad - - - - - - - - \quad (ii)$
Substitute (ii) in (i)

$$\frac{d^2\psi(x)}{dx^2} + 8\pi^2 m \frac{[E - V(x+a)]}{h^2} + \psi(x) = 0 \quad - - - - - - - (iii)$$

- From the block the solution of equation (iii) can be written as multiplication of plane wave & periodicity.

$\psi(x) = u_k(x)e^{ikx} \quad - - - - - - - - - - - \quad (iv)$

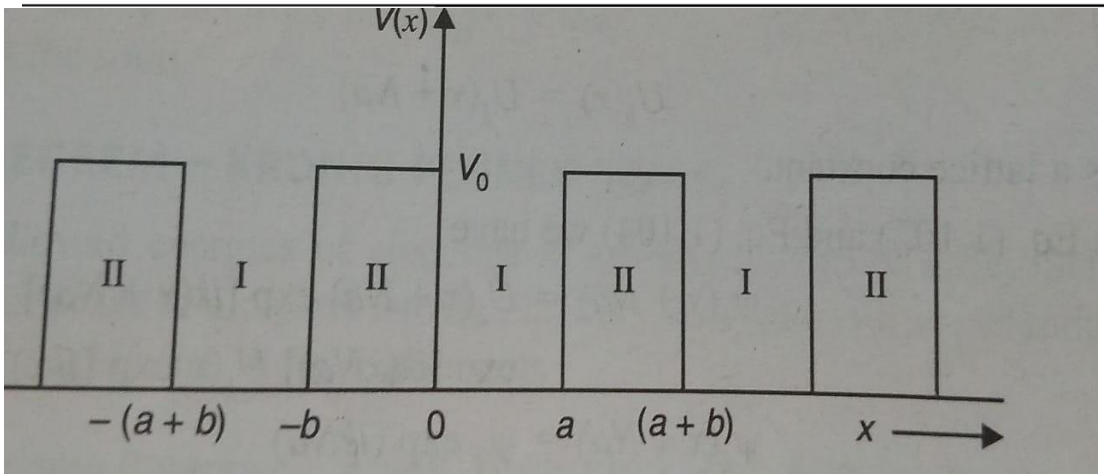
- If we consider 'N' no. of atom then potential

$V(x) = V(x+Na)$
$u_k(x) = u_k(x+Na)$---------- v
Then the solution from bloch.
$\psi(x+Na) = u_k(x+Na)e^{ik}\,(x+Na)$
From equation number "v"
$\psi(x+Na) = [u_k(x)]e^{iKx} \cdot e^{iKNa}$
$= u_k x \cdot e^{iKx} \cdot e^{iKNa} \implies from\ equation(iv)$
$\psi(x+Na) = \psi(x) \cdot e^{iKNa} \quad - - - - - - - - - - - - \quad (vi)$
This is known as Bloch equation (or) theorem complex
Conjugate of equation (vi)
$\psi^*\,(x+Na) = \psi^* \cdot e^{-iKNa} \quad - - - - - - - - - \quad (vii)$
Multiply (vi) & (vii)
$\psi(x+Na) \cdot \psi^*(x) = \psi(x)\psi^*(x)e^{-iKNa}$
$|\psi(x+Na)|^2 = |\psi(x)|^2$
i.e., probability density for electron is equal to entire crystal.

4.Explain the Kronig-Penny model for the electron in periodic potential?

Kronig & penney model:

They proposed the periodic potential.
They approximated a periodicity to a ID rectangular potential with the periodicity (a+b).



) 1D periodic potential function of Kronig-Penny Model

The motion of electron is given by schrodinger wave equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m[E - V(x)]}{h^2}\psi(x) = 0 \quad -------- \quad (i)$$

Potential:

In region-I $V(x) = 0 \qquad 0 < x < a$
region-II $V(x) = V_0 \qquad -b < x < 0$
By applying Schrodinger equation in region-I.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m[E - 0]}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 mE}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \alpha^2\psi(x) = 0 \quad ---------- \quad (ii)$$

Apply Schrodinger wave equation region-II.

$V(x) = V_0, \qquad -b < x < 0$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m[E - V_0]}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m[V_0 - E]}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} - \beta^2\psi(x) = 0 \quad ---------- \quad (iii)$$

$$\beta^2 = \frac{8\pi m(V_0 - E)}{h^2}$$

The solution of equation (ii) & equation (iii) from Bloch can be written as

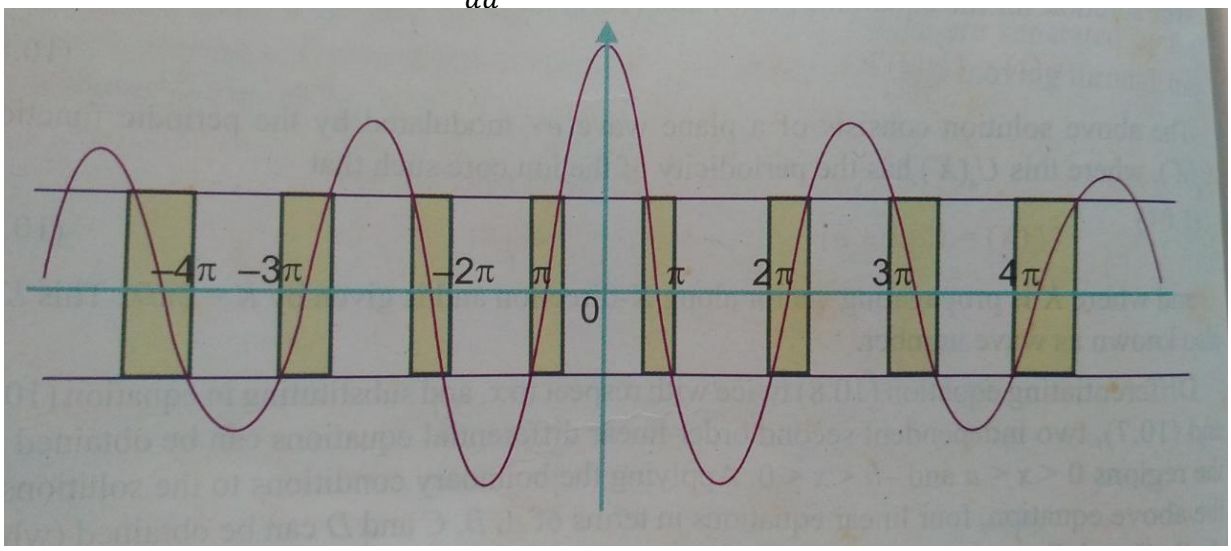
$$\psi(x) = u_k(x)e^{iKx} \quad ---------- \quad (iv)$$

after applying boundary conditions by mathematical expression written as

$$\frac{PSin\alpha a}{\alpha a} + Cos\alpha a = Coska \quad ---------- \quad (v)$$

p➜ Scattering power (or) Barrier strength.

By plotting graph between $\frac{PSin\alpha a}{\alpha a} + Cos\alpha a$ and αa then graph will be



- From the graph electron characterized in an energy level corresponding to particular value of αa. This leads the existence of allowed region and forbidden region decreases.
- If P is very large (P➔∞) P is barrier strength.

$$from\ euqation\ (vii)\ \frac{PSin\alpha a}{\alpha a} + Cos\alpha a = Coska$$

$$\frac{Sin\alpha a}{\alpha a} = 0 \qquad [\therefore P \rightarrow \infty$$

$$Sin\alpha a = 0$$

$$\alpha a = Sin^{-1}(0)$$

$$\alpha a = Sin^{-1}(Sinn\pi)$$

$$\alpha a = n\pi$$

$$\alpha = \frac{n\pi}{a}$$

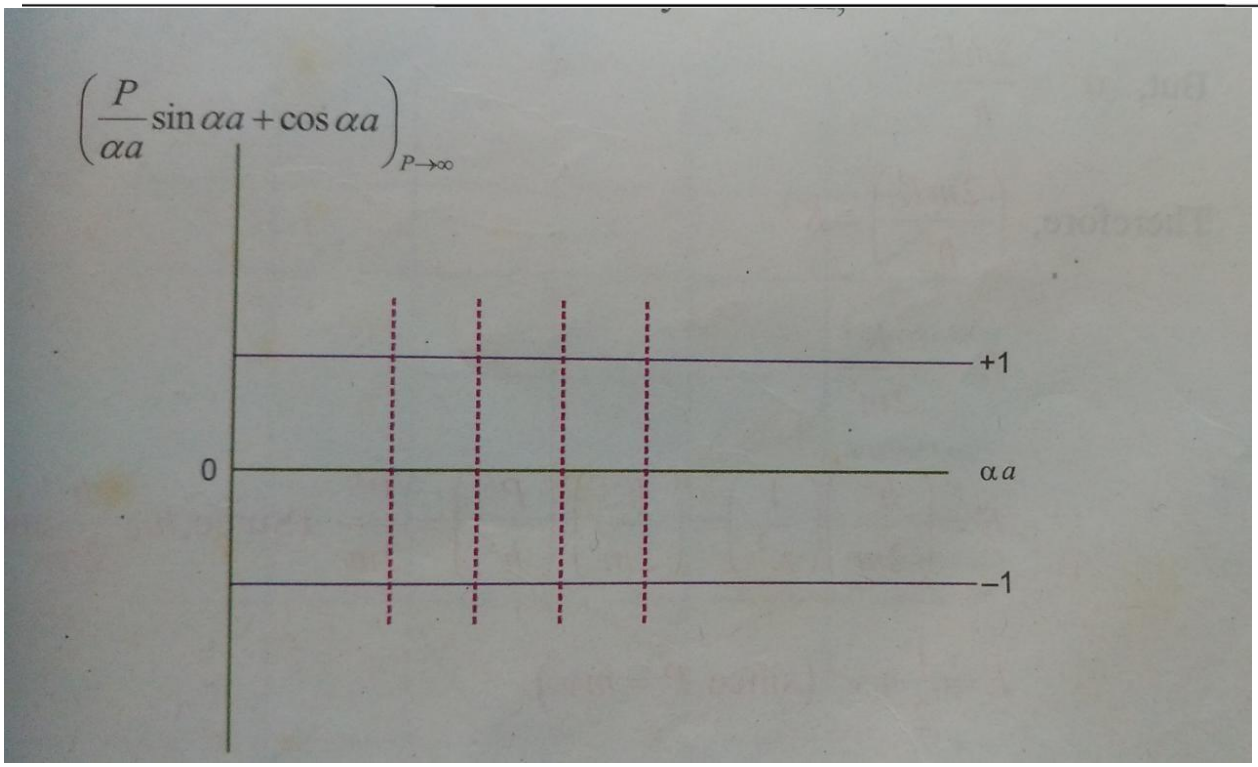
$$\Longrightarrow \alpha^2 = \frac{n^2\pi^2}{a^2}$$

$$\frac{8\pi^2 mE}{h^2} = \frac{n^2\pi^2}{a^2}$$

$$E = \frac{n^2h^2}{8ma^2}$$

i.e., energy is quantized for when barrier strength is more.

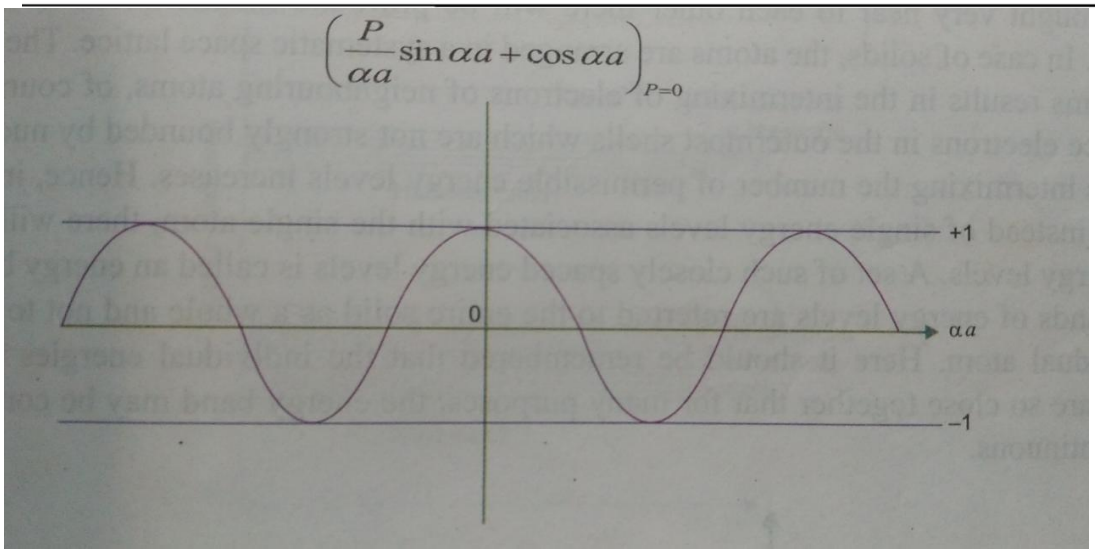
- with increase in barrier strength (P) width of allowed region decreases.



- P is very small P➜0

$$\frac{PSin\alpha a}{\alpha a}+Cos\alpha a=Coska$$
$$Cos\alpha a=Coska$$
$$\alpha a=ka \qquad [\because P=0]$$
$$\alpha=k$$
$$\alpha^2=k^2$$
$$\frac{8\pi^2 mE}{h^2}=\frac{4\pi^2}{\lambda^2} \qquad [\therefore k=\frac{2\pi}{\lambda}\ ]$$
$$\frac{2mE}{h^2}=\frac{1}{\lambda^2} \qquad [\therefore \lambda=\frac{h}{mv}]$$
$$\frac{2mE}{h^2}={}^{1}/_{h^2/m^2v^2}$$
$$\frac{2mE}{h^2}=\frac{m^2v^2}{h^2}$$
$$E=\frac{1}{2}mv^2$$

i.e., the total energy of the electron is from the kinetic energy, potential energy is zero then the particle is free particle then the energy spectrum.



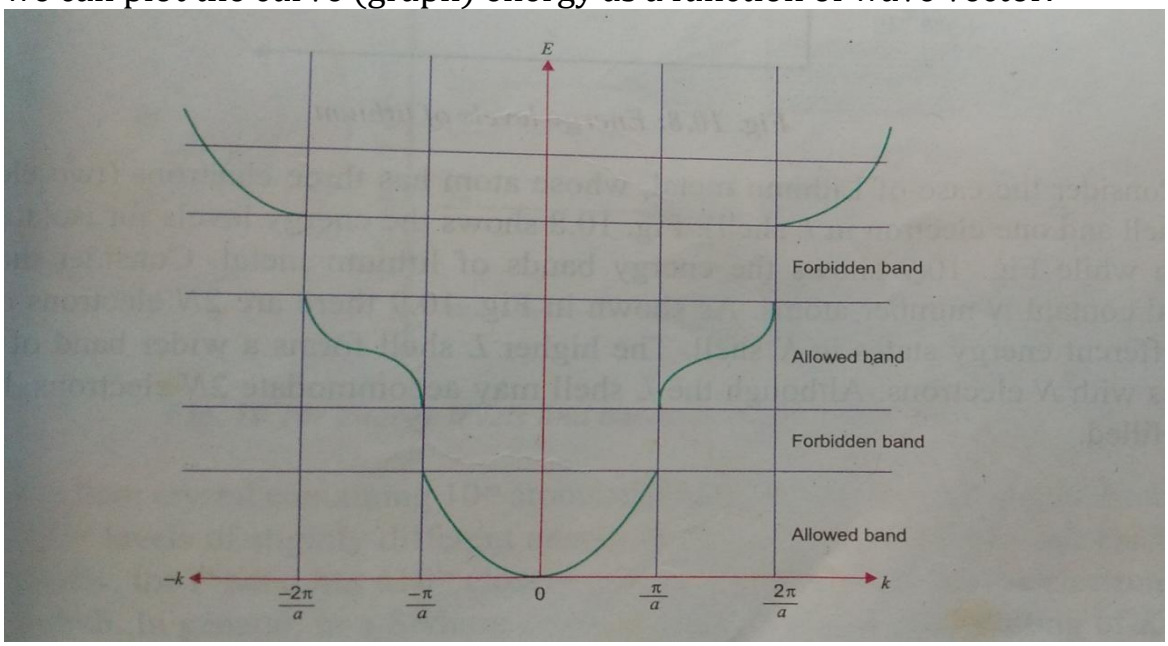
i.e., electron is allowed in all energy band this support classical free electron theory.

5.Draw the E-K Diagram and Define the Brillioun Zone and forbidden zone?

$$\boldsymbol{E-k\ diagram}$$

$E-k\ diagram\ by\ using \frac{PSin\alpha a}{\alpha a}+Cos\alpha a=Coska$

we can plot the curve (graph) energy as a function of wave vector.



In above equation right hand side term ($Coska = \pm 1$) for the values of

$$k=\frac{\pm h\pi}{a}$$

Hence, discontinuity occurs in energy spectrum consist of allowed region and forbidden region.

- If n= 1, $k=\frac{\pm h\pi}{a}$ then allowed region extended $-\frac{\pi}{a}\ to\ \frac{\pi}{a}$ called first Brilliouin zone.
- If n= 2, $k=\frac{\pm 2\pi}{a}$

$i.e., k = -\frac{2\pi}{a}, \frac{-\pi}{a}, 0 \; and \; \frac{\pi}{a}, \frac{2\pi}{a}$ *called second Brilliouin zone.*

Forbidden Energy Band Gap in E–k Diagram

In an E–k diagram, the forbidden energy band gap is a vertical energy range between two bands where no electronic states exist. It appears between the top of the valence band and the bottom of the conduction band. The y-axis (energy E) shows this gap clearly as a region where there are no allowed energy levels for electrons. The gap often occurs at the Brillouin zone boundary (e.g., at k = ±π/a) due to the periodic potential of the crystal.

6.Give the brief explanation about conductors, semiconductors and insulators?

Conductors:

The valence band and conduction bands are overlapped with each other so energy gap (Eg) is zero.

At room temperature free electrons already exist in conduction band. Hence solids (metals) are electrical as well as thermal conductors.

·The electrical resistivity increases with increasing temperature in conductors.

Semiconductors:

The valence band and conduction bands are separated with a small energy band gap (Eg=lev).

At low temperature (0K) free electrons are not available in conduction band hence they like an insulator.

The electrical conductivity increases with increasing temperature thus the semiconductors behave as electrical conductors at high temperature.

Example: Silicon Eg = 1.1eV

Germanium Eg = 0.7eV

Insulators:

The valence band & conduction bands are separated by a very large energy gap (Eg greater or equals to 3eV)

At room temp conduction band is empty & valence band is full of electrons. Even at high temperature valence electrons unable to jump into conduction band.

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UNIT III

Short Question & Answers:

1. Give the Introduction of Quantum Computing?

Quantum computing is an advanced type of computing that uses the principles of quantum mechanics to process information. Unlike classical computers, which use bits (0 or 1), quantum computers use quantum bits (qubits).

Qubits can exist in superposition (0 and 1 at the same time) and can be linked through entanglement, allowing quantum computers to perform many calculations simultaneously. Because of this, quantum computers can solve certain complex problems much faster than classical computers.

2. Write the notation of Bra-Ket?

A Ket is written as $|\psi\rangle$
It represents the state of a quantum system.

- A Bra is written as $\langle\psi|$ It is the complex conjugate transpose of the ket.
- The inner product (Bra–Ket) is written as: $\langle\phi|\psi\rangle$
  It gives the probability amplitude.
- The outer product is written as: $|\psi\rangle\langle\phi|$
  It represents an operator.
- This notation is widely used in quantum mechanics and quantum computing.

3. Write two advantages of Quantum computing?

Very fast processing for complex calculations.

- Can handle many possibilities at the same time.
- Useful for breaking and creating secure codes (cryptography).
- Helps in drug discovery and material research

4. Explain classical logic gates NOR, NAND with truth tables.

NOR gate

The NOR, or NOT OR, gate is a combination OR gate followed by an inverter. Its output is true or high if both inputs are false or low (logic 0).

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

NAND gate

The negated AND, or NAND, gate operates as an AND gate followed by a NOT gate. Its symbol is an AND gate with the circle of a NOT gate at the output. The NAND gate produces a false or low output if both inputs are true.

Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0

5.Explain classical logic gates AND, OR with truth tables

AND gate

The AND gate produces an output of true only when both inputs are true. Otherwise, the output is false.

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR gate

In the OR gate, which is among the most widely used logic circuits in digital devices and components, the output is true if one or both inputs are true. If both inputs are false, the output is false.

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

6.Explain classical logic gates XOR, XNOR with truth tables

XOR gate

The XOR, or exclusive OR, gate acts in the same way as the logical *either/or*. The output is true if either input is true, but not both. The output is false if both inputs are false or both inputs are true.

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

XNOR gate

The XNOR, or exclusive NOR, gate is a combination of an XOR gate followed by an inverter. Its output is true if the inputs are the same and false if the inputs are different.

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1

7.Write the differences of Classical & quantum bits?

Classical Bit	Quantum Bit (Qubit)
Can be 0 or 1 only	Can be 0, 1, or both at the same time
Follows classical physics	Follows quantum mechanics
Cannot exist in superposition	Can exist in superposition
Bits work independently	Qubits can be entangled
Less powerful for complex problems	More powerful for complex problems

Long Question & Answers:

1.Give brief explanation about Superposition?

- Superposition is a basic principle of quantum mechanics.
- It means a quantum Bit an exist in multiple states at the same time.
- In classical computing, a bit is either 0 or 1.
- In quantum computing, a qubit can be 0, 1, or a combination of both simultaneously.

- Mathematically, a qubit in superposition is written as:
  $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$
  - Here, $\alpha$and $\beta$are probability amplitudes.
- When the qubit is measured, the superposition collapses into either 0 or 1.
- Superposition allows quantum computers to process many possibilities at the same time.
- This property gives quantum computers their high computational power compared to classical computers.

2.Give brief explanation about Entanglement?

- Entanglement is an important principle of quantum mechanics.
- It occurs when two or more qubits become strongly correlated with each other.
- In an entangled state, the state of one qubit depends on the state of the other, even if they are far apart.
- Measuring one qubit instantly determines the state of the other qubit.
- The individual qubits cannot be described independently.
- Entanglement has no classical equivalent.
- It is a key resource in quantum computing and quantum communication.
- Applications include quantum teleportation, quantum cryptography, and faster computation.

Example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle+|11\rangle)|$$

This shows two qubits in a maximally entangled state

3.Write the matrix representations of Hadamard (H) gate

Hadamard Gate (H-gate) – It is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gate and its symbol is

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Action of H gate on inputs:

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|0\rangle$ state

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|1\rangle$ state.

$$H|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$H|0\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

If a superposed qubit goes through H gate, the result will be

$$H|\psi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}}\right)|0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right)|1\rangle$$

$$H|\psi\rangle = \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Gate representation is

$$\alpha|0\rangle + \beta|1\rangle \text{ ——— } \boxed{H} \text{ ——— } \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

The truth table is as follows

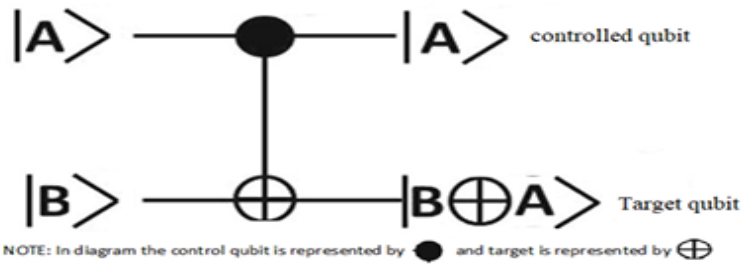
Input	**Output**
$\|0\rangle$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$
$\|1\rangle$	$\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\left(\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}\right)$

4. Explain the working of the CNOT (Controlled-NOT) gate with its circuit and matrix form.

1. Controlled Gate (CNOT)

The CNOT gate is a two-qubit operation, where the first qubit is referred as the control qubit $|A\rangle$ and the second qubit as the target qubit $|B\rangle$. If the control qubit is $|1\rangle$ then it will flip the target qubit state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$. When the control qubit is in state $|0\rangle$ then the target qubit remains unchanged.

The symbolic representation is as follows. The upper line represents control qubit and bottom line represents target qubit.



In the combined qubit, first term is control qubit and the second term is target qubit. For ex, in $|AB\rangle$, A is control qubit and B is target qubit

Matrix form of **CNOT** Gate is given by

$$\mathbf{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The Transformation could be expressed as $|A, B\rangle \rightarrow |A, B \oplus A\rangle$

Action of the gate: Consider the operations of CNOT gate on the four inputs $|00\rangle$,$|01\rangle$,$|10\rangle$ and $|11\rangle$.

i) Operation of CNOT Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$

Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|0\rangle$.

ii) Operation of CNOT Gate for input $|01\rangle$: $|01\rangle \rightarrow |01\rangle$

Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|1\rangle$

iii) Operation of CNOT Gate for input $|10\rangle$: $|10\rangle \rightarrow |11\rangle$

Here, control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|0\rangle$ to $|1\rangle$.

iv) Operation of CNOT Gate for input $|11\rangle$: $|11\rangle \rightarrow |10\rangle$

Here control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|1\rangle$ to $|0\rangle$.

The Truth Table of operation of CNOT gate is as follows.

Input	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|11\rangle$
$\|11\rangle$	$\|10\rangle$

5. Write the matrix representations of Pauli-X, Pauli-Y, and Pauli-Z gates

Single qubit gates

1. X – Gate

This is also called as Pauli X – gate. It is given by

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

When X operates on $|0\rangle$ and $|1\rangle$ the output will be inverted (ie, $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes$|0\rangle$)

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Since X inverts each input it is also called as *bit-flip* gate. If a superposed qubit goes through X gate, the result will be

$$X|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

So,

$$X|\Psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

Symbolically these gates are represented as follows

2. Y – Gate

This is also called as Pauli Y – gate. It is given by

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

When Y operates on $|0\rangle$ and $|1\rangle$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \times 1 + (-i) \times 0 \\ i \times 1 + 0 \times 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i\begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \times 0 + (-i) \times 1 \\ i \times 0 + 0 \times 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i\begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

So,

$$Y|0\rangle = i|1\rangle \text{ and } \qquad Y|1\rangle = -i|0\rangle$$

If a superposed qubit goes through Y gate, the result will be

$$Y|\Psi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \times \alpha + (-i) \times \beta \\ i \times \alpha + 0 \times \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i\beta|0\rangle + i\alpha|1\rangle$$

So,

$$Y|\Psi\rangle = i\alpha|1\rangle - i\beta|0\rangle$$

Symbolically these gates are represented as follows

3. Z – Gate

This is also called as Pauli Z – gate. It is given by

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

When Z operates on $|0\rangle$ and $|1\rangle$ the phase will change. Hence this is also called as *phase-flip* gate

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 0 \times 0 \\ 0 \times 1 + (-1) \times 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 0 \times 1 \\ 0 \times 0 + (-1) \times 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -|1\rangle$$

So,

$$Z|0\rangle = |0\rangle \text{ and } Z|1\rangle = -|1\rangle$$

If a superposed qubit goes through Z gate, the result will be

$$Z|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 \times \alpha + 0 \times \beta \\ 0 \times \alpha + (-1) \times \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

So,

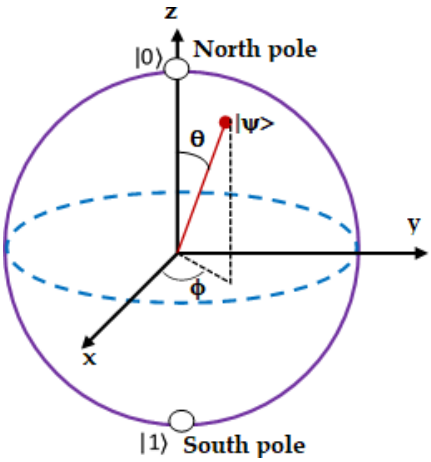
$$Z|\Psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

Symbolically these gates are represented as follows

6.Explain the Bloch sphere representation of a qubit with a neat diagram.

Bloch sphere

Bloch sphere is an imaginary sphere which is used to represent pure single-qubit states as a point on its surface. It has unit radius. Its North Pole and South Pole are selected to represent the basis states namely $|0\rangle$ and $|1\rangle$. North Pole represents $|0\rangle$ (say spin up↑) and South Pole represents $|1\rangle$ (say spin down ↓). All other points on the sphere represent superposed states (ie, state space). Bloch sphere allows the state of a qubit to be represented in spherical coordinates (ie, r, θ and ϕ). It is as follows



The state qubit $|\psi\rangle$ on the Bloch sphere makes an angle θ with z-axis and its projection (azimuth) makes angle ϕ with x-axis as shown. It is clear from the fig that $0 < \theta < \pi$ and $0 < \phi < 2\pi$. $|\psi\rangle$ is represented as

$$|\psi\rangle = \alpha\,|0\rangle + \beta|1\rangle$$

It can be proved that

$$|\Psi\rangle = cos\frac{\theta}{2}|0\rangle + e^{i\phi}\,sin\frac{\theta}{2}|1\rangle - - - (1)$$

Using this equation we can represent $|\psi\rangle$ for different θ and ϕ as follows

Case-1: let $\theta = 0$ and $\phi = 0$, then eq (1) becomes

$$|\Psi\rangle = cos\ 0|0\rangle + e^{i0}sin0|1\rangle = |0\rangle + 0$$

$$\therefore |\Psi\rangle = |0\rangle$$

This is the **north pole**, representing the base state $|0\rangle$.

Case-2: let $\theta = \pi$ and $\phi = 0$, then eq (1) becomes

$$|\Psi\rangle = cos\ \frac{\pi}{2}|0\rangle + e^{i0}sin\frac{\pi}{2}|1\rangle = 0 + |1\rangle$$

$$\therefore |\Psi\rangle = |1\rangle$$

This is the **south pole**, representing the base state $|1\rangle$.

Case-3: let $\theta = \pi/2$ and $\phi = 0$, then eq (1) becomes

$$|\Psi\rangle = \cos\frac{\pi}{4}|0\rangle + e^{i0}\sin\frac{\pi}{4}|1\rangle$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\Psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

This point lies on the **positive x-axis**, representing the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Case-4: let $\theta = \pi/2$ and $\phi = \pi$, then eq (1) becomes

$$|\Psi\rangle = \cos\frac{\pi}{4}|0\rangle + e^{i\pi}\sin\frac{\pi}{4}|1\rangle$$

$$|\Psi\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$|\Psi\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

This point lies on the **negative x-axis**, representing the superposition $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

Unit IV

Short Question & Answers:

1. What is the main difference between classical circuits and quantum circuits

Classical Circuits	Quantum Circuits
① Basic unit of Classical is Bit (0 or 1)	① Basic unit of Quantum is Qubit [Superposition of 0 and 1]
② State representation of Classical single definite value (0 or 1)	② State representation of Quantu is vector in Hilbert Space Eg: $\alpha\|0\rangle + \beta\|1\rangle$
③ Classical operate based on Boolean algebra and deterministic logic gates AND, OR etc.	③ Quantum linear algebra matrices, tensors, vectors, unitary operators.
④ Classical most gates are irreversible	④ Quantum All gates must be reversible.

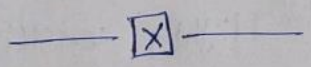
2. What are the basic notations used in quantum circuits, such as |0⟩, |1⟩, and gate symbols?

Basic circuit notation :-

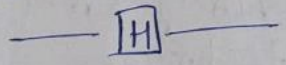
A quantum circuit is a diagramatic representation of how qubits are processed in a quantum computation.

The basic notations used in circuits so that operations can be easily understood.

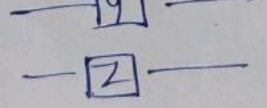
→ X gate : Flips the qubit



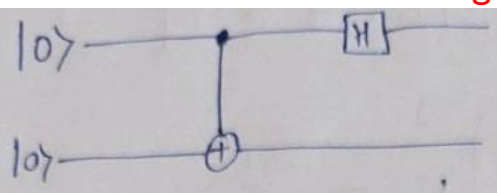
→ H gate : Creates superposition.



→ Z and Y gates: Apply phase and rotation changes.



3. Draw the circuit for non-entanglement



A CNOT gate is applied to the |00⟩ state followed by a Hadamard gate on the first qubit.

$\psi_i = |00\rangle$

$\psi_f = (H \otimes I)\ CNOT\ |00\rangle$

4. Explain the Introduction to quantum circuits

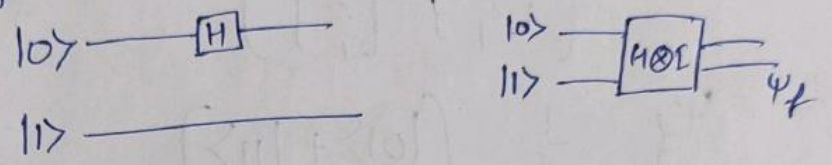
Introduction to Quantum Circuits:-

A Quantum circuit is a sequence of Quantum gates applied to a set of Qubits. Qubits can exist in a superposition of both states simultaneously.

This property along with Entaglement and Interference, allows quantum circuits to perform complex computations more efficiently than classical circuits for certain problems.

5. Draw the circuit diagram to create a superposition using H and NOP

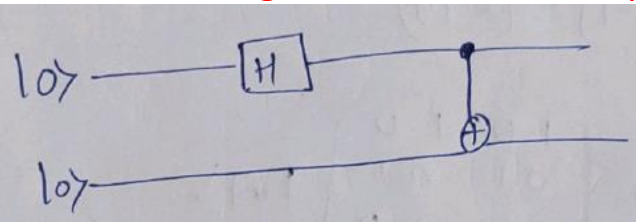
circuit diagram.



$$\psi_f = (H \otimes I)(|0\rangle \otimes |1\rangle)$$

$$\psi_f = (H \otimes I)|01\rangle$$

6. Draw the circuit diagram to create a Bell pair using H and CNOT gates.



$$\psi_f = |00\rangle$$

A Hadamard gate is applied to the first qubit, followed by a CNOT gate.

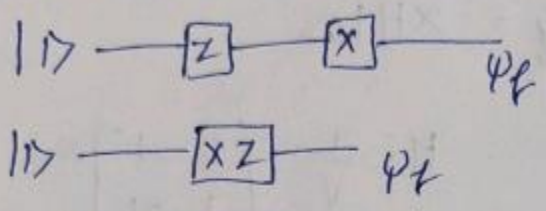
$$\psi_f = CNOT\,(H \otimes I)|00\rangle$$

Long Question & Answers:

1. Find the final qubit state for the circuit ——[Z]——[X]—— Using inputs |0⟩ and |1⟩

Apply Z gate followed by X gate on $|1\rangle$

Consider the circuit



$\psi_i = |1\rangle$, $\psi_f = XZ|1\rangle$

Matrices are

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

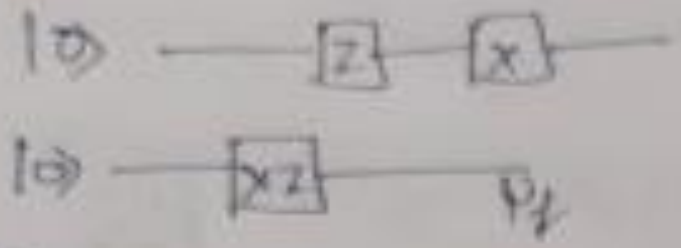
$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$XZ|1\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -|0\rangle$$

$$\psi_f = XZ|1\rangle$$

$$\psi_f = -|0\rangle$$

Z – X on $|0\rangle$



$\psi_i = |0\rangle \quad \psi_f = XZ|0\rangle$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

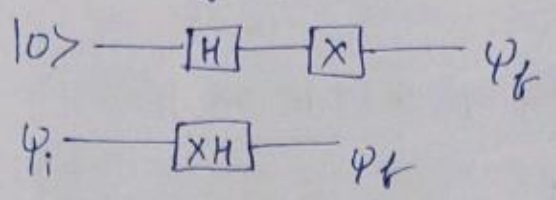
$$XZ|0\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$\psi_f = XZ|0\rangle = |1\rangle$$

2. Find the final qubit state for the circuit ——[H]——[X]— Using inputs $|0\rangle$ and $|1\rangle$

Apply Hadamard (H) gate followed by X gate on $|0\rangle$

Circuit diagram.



$\psi_i = |0\rangle$, $\psi_f = XH|0\rangle$

Matrices $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$XH = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$XH|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\left[\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right]$$

$$= \frac{1}{\sqrt{2}}\left[|0\rangle + |1\rangle\right]$$

$$\psi_f = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

H Creates a superposition and X flips the components but keeps the superposition same.

H–X on $|1\rangle$

$|1\rangle$ —— $\boxed{H}$ —— $\boxed{X}$ ——

$\psi_i|1\rangle \Rightarrow \psi_f = XH|1\rangle$

$$XH|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

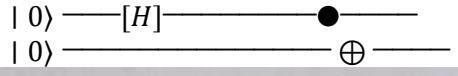
$$XH|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

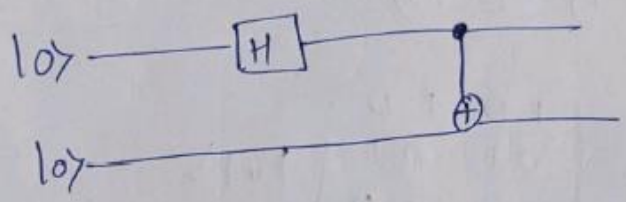
$$= \frac{1}{\sqrt{2}}\left[-\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

$$= \frac{1}{\sqrt{2}}\left[-|0\rangle + |1\rangle\right]$$

3. Find the final state of the following circuit:



Applying H gate and CNOT gate



$\psi_i = |00\rangle$

A Hadamard gate is applied to the first qubit, followed by a CNOT gate.

$$\psi_f = CNOT\,(H \otimes I)\,|00\rangle$$

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\varphi_f = \frac{1}{\sqrt{2}}\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}\begin{bmatrix}1&0&1&0\\0&1&0&1\\1&0&-1&0\\0&1&0&-1\end{bmatrix}|00\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}1&0&1&0\\0&1&0&1\\0&1&0&-1\\1&0&-1&0\end{bmatrix}|00\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}1&0&1&0\\0&1&0&1\\0&1&0&-1\\1&0&-1&0\end{bmatrix}\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

multiple with $|00\rangle$. This operation Pull out the first column of the matrix

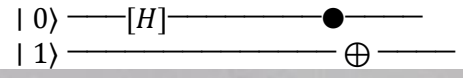
$$\varphi_f = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

$$\varphi_f = \frac{1}{\sqrt{2}}\left[\begin{bmatrix}1\\0\\0\\0\end{bmatrix}+\begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right]$$

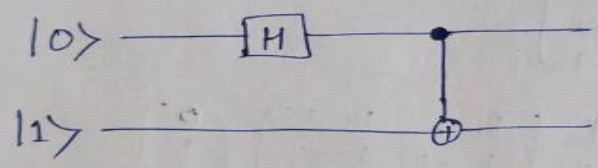
$$\varphi_f = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

This is the Entangled State or Bell state.

4. Find the final state of the following circuit:



Entanglement using Quantum Circuits :-



The system starts with two separate qubits

→ A Hadamard gate (H) is applied to the first qubit $|0\rangle$

→ The second qubit $|1\rangle$ is unchanged.

$\psi_i = |01\rangle$

$\psi_f = CNOT\,(H \otimes I)\,|01\rangle$

$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$$\psi_f = CNOT\,\frac{1}{\sqrt{2}}\left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)|01\rangle$$

$$= CNOT\,\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}|01\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}|01\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}\right)$$

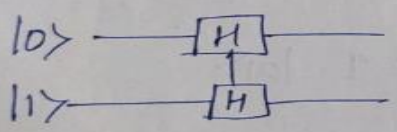
$$\psi_f = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)$$

This is a Entangled State known as a Bell State.

5. Find the final state of the following circuit:

$|0\rangle$ ——[H]——
$|1\rangle$ ——[H]——

Applying a Hadamard (H) gate to Each qubit in a 2-qubit System



A Hadamard gate is applied to the first qubit, and another H-gate is applied to the second qubit. This parallel operation is represented by the tensor product $(H \otimes H)$

$\psi_i = |01\rangle$

$\psi_f = (H \otimes H)\,|01\rangle$

Using matrix mechanics

$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$H \otimes H = \left(\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\right) \otimes \left(\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\right)$

$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$|01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

$|11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$= \frac{1}{2}\begin{bmatrix} 1\times\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & 1\times\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ 1\times\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & -1\times\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \end{bmatrix}$

$= \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$

$(H \otimes H)\,|01\rangle = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

The vector $|01\rangle$ only has a '1' in the second position, this operation simply pulls out the second column of the matrix.

$$\psi_f = \frac{1}{2}\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix}$$

$$\psi_f = \frac{1}{2}\left[\begin{bmatrix}1\\0\\0\\0\end{bmatrix} - \begin{bmatrix}0\\1\\0\\0\end{bmatrix} + \begin{bmatrix}0\\0\\1\\0\end{bmatrix} - \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right]$$

Converting each vector back to its ket representation

$$\psi_f = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

So both qubits together are in superposition of four states. $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$

6. Find the final state of the following circuit:

```
|0⟩ ——[ H ]———
|1⟩ ——————————
```

Superposition using Quantum Circuit :-

Superposition is a fundamental property of quantum mechanics where a qubit can exist in a combination of basis states $|0\rangle$ and $|1\rangle$ simultaneously.

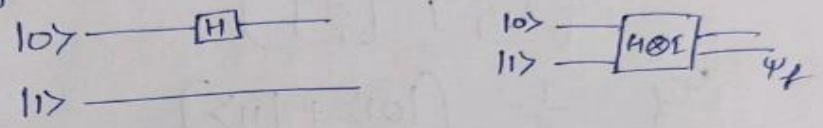
A Quantum Circuit Can be used to Create Superposition using the Hadmard gate (H gate).

Consider a 2 Qubit system Initially in the state

$|\psi_i\rangle = |01\rangle$

A Hadamard gate is applied only on the first Qubit and second Qubit is left unchanged.

Circuit diagram.



$$\psi_f = (H \otimes I)(|0\rangle \otimes |1\rangle)$$

$$\psi_f = (H \otimes I)|01\rangle$$

Matrics $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$H \otimes I|01\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes I \; |01\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1\times I & 1\times I \\ 1\times I & -1\times I \end{bmatrix}|01\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}|01\rangle$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right)$$

$$\psi_f = \frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right)$$

So both Qubit together are in a superposition of two states $|01\rangle$ and $|11\rangle$

UNIT V:

Short Question & Answers:

1. Write two applications of quantum computing in cybersecurity?

Quantum Key Distribution (QKD):

- If a hacker tries to intercept the key, the quantum state changes automatically and is detected.
- Used for highly secure government, military, and banking communication.

Quantum Random Number Generation (QRNG):

- Secure encryption requires strong random numbers.
- This protects passwords, encryption keys, and financial transactions.

2. Mention any two applications of quantum deep learning?

Quantum Neural Networks (QNNs):

- Quantum circuits can work like neural networks but can represent more complex patterns.
- QNNs use qubits and quantum gates to learn data efficiently.

Handling High-Dimensional Data

- Quantum states naturally exist in extremely large-dimensional spaces.
- This helps deep learning models understand complex data with many features.

3. Write two ethical applications/concerns in quantum computing.

Breaking Current Encryption:

- Quantum computers can solve mathematical problems extremely fast.
  Because of this:
- They can break today's widely used internet security systems like RSA and ECC.

Privacy Risks:

- If quantum computers are used wrongly:
- Personal data like Aadhaar details, bank details, medical records may be accessed.

4. Mention two applications of quantum artificial intelligence?

Better Search Algorithms:

- Quantum algorithms like Grover's algorithm can search large databases much faster.
- This helps AI systems quickly find important information from huge data.

Big Data Analytics:

- Quantum AI can handle extremely large datasets that classical computers struggle with.

- This supports faster trend prediction in business, healthcare, and finance.

5. Write two applications of quantum computing in molecular modelling?

Battery Material Design:

- Quantum computers can predict how lithium ions move inside battery materials.
- This helps design batteries that charge faster, last longer, and don't heat up.

Solar Cell Improvement:

- Quantum computers study how light creates electrons inside solar materials.
- This helps discover new materials that absorb more sunlight and increase solar efficiency.

6. Describe any two applications of Quantum Neural Networks (QNNs) in real-world problem-solving?

Medical Diagnosis:

- QNNs can study medical scans like **MRI, CT, X-ray** much faster.
- This helps doctors take **quick and correct decisions.**

Weather Prediction:

- Weather prediction needs huge data: temperature, wind, air pressure, oceans, etc.
- This leads to **more accurate forecasts** and early warnings for floods and cyclones.

Long Question & Answers:

1. Explain the applications of quantum computing in artificial intelligence

Better Search Algorithms:

- Quantum algorithms like Grover's algorithm can search large databases much faster.
- This helps AI systems quickly find important information from huge data.
- It improves the speed of recommendations, data retrieval, and decision-making.

Big Data Analytics:

- Quantum AI can handle extremely large datasets that classical computers struggle with.
- It can analyze millions of records at once and detect hidden patterns.
- This supports faster trend prediction in business, healthcare, and finance.

Quantum Robots:

- Robots with quantum processors can analyze many possible actions at the same time.
- They can make quick and smart decisions in complex environments like rescue missions or factories.
- This improves their accuracy, safety, and intelligence.

Predictive Healthcare:

- Quantum AI can examine large medical datasets and identify disease patterns very early.
- It helps doctors detect serious diseases like cancer at an early stage.
- This allows better treatment planning and preventive care.

Real-Time Language Translation:

- Quantum AI can process multiple language patterns simultaneously.
- This makes translation faster and more accurate in real time.
- It helps with global communication, tourism, and international business.

2. Explain the major applications of quantum computing in molecular modelling and new material design.

Drug Discovery:

- Quantum computers can study how a drug molecule attaches to a protein in the body.
- They help find the best drug shape, reduce side effects, and make medicine development faster.

Battery Material Design:

- Quantum computers can predict how lithium ions move inside battery materials.
- This helps design batteries that charge faster, last longer, and don't heat up.

Solar Cell Improvement:

- Quantum computers study how light creates electrons inside solar materials.
- This helps discover new materials that absorb more sunlight and increase solar efficiency.

Carbon Capture Materials:

- Quantum computers model how CO_2 sticks to porous materials.
- This helps design materials that absorb more CO_2 and reduce air pollution.

Aerospace Alloys:

- Quantum computers study how atoms combine in metals.
- This helps create lighter and stronger alloys for airplanes and spacecraft.

3. Explain the role of quantum computing in improving cybersecurity applications?

Quantum Key Distribution (QKD):

- QKD uses quantum particles (photons) to share secret keys between two users.
- If a hacker tries to intercept the key, the quantum state changes automatically and is detected.
- This makes communication channels almost impossible to hack.
- Used for highly secure government, military, and banking communication.

Quantum Random Number Generation (QRNG):

- Secure encryption requires strong random numbers.
- Quantum computers use unpredictable quantum behaviour to generate true randomness.
- This protects passwords, encryption keys, and financial transactions.
- Much safer than classical random number generators.

Fast Detection of Cyber Attacks:

- Quantum machine learning can analyze enormous network traffic in seconds.
- It quickly identifies unusual patterns like malware, intrusions, and data breaches.
- This allows organizations to respond to attacks much faster than classical tools.
- Useful for banks, cloud companies, and large networks.

Secure Authentication Systems:

- Quantum properties like superposition and entanglement can be used to create identity signatures.
- These "quantum fingerprints" cannot be copied or forged.
- This makes login systems and biometric devices extremely secure.
- Useful for airports, defense, and secure data centers.

Protecting IoT Devices:

- Quantum cryptography can secure millions of IoT devices like sensors, cameras, and smart home devices.
- Because IoT devices use weak passwords, quantum-safe encryption gives them stronger protection.
- Prevents hacking of smart homes, vehicles, medical devices, and industrial machines.

4. Discuss the applications of quantum computing in deep learning?

Quantum Neural Networks (QNNs):

- Quantum circuits can work like neural networks but can represent more complex patterns.
- QNNs use qubits and quantum gates to learn data efficiently.
- They can solve problems that classical neural networks find difficult, especially high-dimensional tasks.

Handling High-Dimensional Data

- Quantum states naturally exist in extremely large-dimensional spaces.
- This helps deep learning models understand complex data with many features.
- Useful for tasks like physics data, and image/video analysis.

Quantum Speed-up for Data Search

- Algorithms like Grover's search can find important patterns faster than classical search methods.
- This speeds up classification, clustering, and pattern detection in deep learning.
- Useful for big-data applications.

Efficient Feature Extraction

- Quantum circuits can extract useful features from raw data automatically.
- This reduces the number of parameters in a deep learning model.
- The model becomes faster, lighter, and more accurate.

Enhanced Security of Deep Learning Models

- Quantum tools help detect adversarial attacks on neural networks.
- They can build more secure models for finance, healthcare, and defense.
- Quantum cryptography can also be combined with deep learning for complete protection.

5. Describe the applications of Quantum Neural Networks (QNNs) in real-world problem-solving?

Image & Speech Recognition:

- In normal AI, processing large images or voice files is slow.
- But QNNs use quantum states to **compress and analyze** big data quickly.
- So voice assistants, cameras, and medical imaging become **faster and more accurate**.

Medical Diagnosis:

- QNNs can study medical scans like **MRI, CT, X-ray** much faster.
- They can spot tiny changes (tumors, infections) more accurately than classical models because of quantum sensitivity.
- This helps doctors take **quick and correct decisions.**

Weather Prediction:

- Weather prediction needs huge data: temperature, wind, air pressure, oceans, etc.
- QNNs can process this massive data quickly and identify hidden climate patterns.
- This leads to **more accurate forecasts** and early warnings for floods and cyclones.

Autonomous Driving (Self-Driving Cars):

- A self-driving car must make decisions every second — read signals, detect people, avoid accidents.
- QNNs help by processing video, radar, and sensor data in real time.
- This makes driving **safer and smoother**.

Language Understanding:

- QNNs help AI understand human language better — tone, meaning, context.
- This improves **translation apps and voice assistants** like Siri, Alexa, and Google Assistant.

6. Write ethical applications/concerns in quantum computing?

Breaking Current Encryption:

- Quantum computers can solve mathematical problems extremely fast.

Because of this:

- They can break today's widely used internet security systems like RSA and ECC.
- Passwords, bank information, and private messages that are safe now may become unsafe.
- This can cause online hacking, data leakage, and misuse of sensitive information.

Privacy Risks:

- If quantum computers are used wrongly:
- Personal data like Aadhaar details, bank details, medical records may be accessed.
- Hackers can misuse quantum technology to read encrypted messages.
- This increases the chances of identity theft, fraud, and financial loss.
- People may lose trust in digital systems, affecting online safety.

Job Loss:

- Automation becomes faster with quantum computing.
- Traditional IT tasks like data processing, encryption, and optimization may shift to quantum machines.
- Classical computing jobs may reduce in areas such as cybersecurity, network management, and software testing.
- Workers may need to learn new quantum-related skills to stay relevant.

Data Misuse:

- Quantum computers can analyze huge amounts of data quickly.
- Without proper laws, sensitive information may be misused.
- Medical, financial, or personal records can be exploited for profit or manipulation.
- Companies or governments may use data in unethical ways, affecting individual freedom.

Environmental Impact:

- Quantum computers need extremely cold temperatures (near absolute zero).
- Cooling machines consume huge amounts of electricity.
- If energy comes from non-renewable sources, it increases carbon emissions.
- Environment and sustainability must be considered while building quantum labs.